

Laila Mahmoud

Data Science Assurance Associate

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Education

Cairo University

M.Sc. in Data Science

CGPA: 3.86

June 2021, Giza, Egypt

Relevant Courses: Data Mining - Machine Learning - Advanced Analytics - Big Data Technologies

Experience

Data Science Assurance Associate

Ernst & Young Egypt

Full-time - On-Site

Jan 2021 - Present

- Core member of a team developing an automated review platform from scratch for E-discovery, implementing predictive coding and topic modelling to reduce labor costs and time.
- Conducted research and development for classification models, predictive analysis, and mining of information from text data, analyzing outputs and monitoring precision for the entire tool.
- Developed classifier models to identify "red flags" and fraud-related issues, assisting in predictive coding and topic modelling from evidence following EY standards.

Projects

TECHNOLOGY ASSISTED REVIEW (TAR) Platform

- Developed an automated review platform from scratch for E-discovery, implementing predictive coding and topic modelling for efficient reviews.
- Conducted R&D for classification models, predictive analysis, and text data mining; analyzed outputs and monitored precision for the tool.
- Created classifier models to identify "red flags" and fraud-related issues using EY standards, incorporating predictive coding and topic modelling.

TEXT ANALYTICS - Motor Vehicle Customer Review Data

- Performed sentiment (Positive, Negative & Neutral) and time series analysis on customer comments for over a year across four categories.
- Created heat maps of terms by survey category based on word frequency and extracted Positive/Negative word clouds.
- Developed customized Tableau dashboards for effective reporting and visualizations.

CHATBOT

- Developed a user-friendly chatbot for product-related queries (hours of operation, reservation options), providing an overview via a QA platform.
- Engineered chatbot intelligence to build question pipelines based on user requirements and provide relevant/recommended questions.

INFORMATION GOVERNANCE

- Scanned and parsed data from multiple file formats, extracted metadata, pushed results for Elasticsearch indexing, and created interactive Kibana dashboards.
- Performed ROT Analysis (Redundant, Outdated, or Trivial) on data to identify content patterns.
- Executed full-text search analysis on Elasticsearch to tag Personally Identifiable Information (PII) such as social security numbers, addresses, and names.

FRAUD ANALYTIC PLATFORM (FAP)

- Developed FAP, a Fraud Analytics and investigative platform with an inbuilt case manager and a suite of analytics for various ERP systems.
- Enabled clients to interrogate their accounting systems for anomaly identification and fraud indicators by running advanced analytics.

Skills

Programming Languages: Python (pandas, numpy, scipy, scikit-learn, matplotlib), SQL, Java, JavaScript/jQuery

Machine Learning: Regression, SVM, Naïve Bayes, KNN, Random Forest, Decision Trees, Boosting, Cluster Analysis, Word Embedding, Sentiment Analysis, Natural Language Processing, Dimensionality Reduction, Topic Modelling (LDA, NMF), PCA, Neural Nets

Database & Visualizations: MySQL, SQLServer, Cassandra, HBase, Elasticsearch, D3.js, DC.js, Plotly, Kibana, Matplotlib, ggplot, Tableau

Other Technologies: Regular Expression, HTML, CSS, Angular 6, Logstash, Kafka, Python Flask, Git, Docker, Computer Vision (OpenCV), Deep Learning (understanding of)